

**DATA-DRIVEN BANK
FRAUD ANALYTICS AND
RISK MONITORING USING
TRANSACTION AND
CUSTOMER BEHAVIOUR
ANALYSIS**

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ABSTRACT

The rapid growth of digital banking and electronic financial transactions has increased the complexity and risk of fraudulent activities. Banks process a large volume of transactions across different transaction types, merchant categories, devices, locations, account types, and time periods. Identifying potentially fraudulent transactions manually from such large datasets is difficult and inefficient.

This project, titled “Data-Driven Bank Fraud Analytics and Risk Monitoring Using Transaction and Customer Behaviour Analysis,” focuses on analyzing banking transaction data to identify patterns and characteristics associated with fraudulent transactions. The project uses a dataset containing 200,000 banking transactions and 24 original attributes, with `Is_Fraud` serving as the target variable.

The project follows a data analytics workflow consisting of data inspection, data cleaning, feature engineering, exploratory data analysis, SQL-based analysis, and interactive dashboard development. Additional analytical features such as Hour, Age Group, Day Name, Month Name, Transaction Size, Balance Category, Time of Day, Balance Utilization, Risk Score, and Risk Level were created to support deeper analysis.

A rule-based heuristic risk scoring system was developed using predefined business rules involving transaction size, time of day, account balance, and balance utilization. The scoring system categorizes transactions into Low Risk, Medium Risk, and High Risk. This risk score is analyst-defined and is not generated using a machine learning model.

The analysis identified several important fraud patterns. The overall fraud rate was approximately 5.04%, with 10,088 fraudulent transactions out of 200,000 transactions. Mobile transactions showed a higher fraud rate than the other device types, while Late Night transactions showed a substantially higher fraud rate compared with other times of day. High-risk transactions also showed a higher fraud rate than Medium Risk and Low Risk transactions.

An interactive Tableau dashboard was developed to provide monitoring of overall transactions, fraud volume, fraud rate, risk levels, transaction types, device types, merchant categories, states, and time of day. SQL views were also created to support dashboard-oriented reporting.

The project demonstrates how data analytics, SQL, feature engineering, rule-based risk scoring, and business intelligence visualization can be combined to support banking fraud monitoring and data-driven decision-making.

Keywords: Bank Fraud, Fraud Analytics, Transaction Analysis, Customer Behaviour, Risk Monitoring, Data Analytics, SQL, Tableau, Feature Engineering, Rule-Based Risk Scoring.

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The banking industry has experienced significant digital transformation through online banking, mobile banking, electronic payments, ATMs, point-of-sale systems, and other digital transaction channels. While these technologies provide convenience and accessibility to customers, they also create opportunities for fraudulent activities.

Financial institutions handle a large number of transactions every day. These transactions may differ based on transaction amount, transaction type, merchant category, device type, location, account type, customer characteristics, and time of transaction. Fraudulent transactions may exhibit unusual combinations of these characteristics.

Traditional manual monitoring is difficult when transaction volumes are large. Therefore, data analytics can be used to examine transaction records systematically, identify patterns, compare fraudulent and genuine transactions, and support fraud monitoring.

This project applies data analytics techniques to a banking transaction dataset to understand fraudulent transaction behaviour and develop a structured risk-monitoring approach.

1.2 Problem Statement

The increasing volume of digital banking transactions makes manual identification of fraudulent activities difficult. Banks need effective methods to understand transaction patterns and identify characteristics associated with fraudulent transactions.

The problem addressed in this project is to analyze a large banking transaction dataset and determine:

- How fraudulent transactions are distributed across different categories.

- Which transaction types have higher fraud rates.
- Which devices and merchant categories are associated with higher fraud rates.
- Whether fraud varies according to time of day, day of week, state, account type, and age group.
- Whether transaction amount and balance utilization differ between fraudulent and genuine transactions.
- Whether a rule-based risk score can help categorize transactions according to risk.
- How these insights can be presented through an interactive dashboard for fraud monitoring.

1.3 Aim of the Project

The primary aim of this project is to perform data-driven bank fraud analytics and risk monitoring using transaction and customer behaviour data.

The project focuses on identifying meaningful fraud patterns through exploratory analysis, feature engineering, SQL analysis, rule-based risk scoring, and interactive visualization.

1.4 Objectives of the Project

The major objectives of the project are:

1. To analyze customer and transaction behaviour using banking transaction data.
2. To identify patterns associated with fraudulent transactions.
3. To compare fraudulent and genuine transactions based on transaction amount, account balance, transaction type, device type, time, and other characteristics.
4. To calculate fraud rates across different transaction and customer-related categories.
5. To engineer analytical features such as transaction size, age group, time of day, balance utilization, and risk level.

6. To develop a rule-based heuristic risk scoring system for transaction risk monitoring.
7. To analyze fraud patterns using SQL queries and analytical views.
8. To develop an interactive Tableau dashboard for monitoring fraud trends and risk patterns.
9. To generate meaningful insights that can support banking fraud-monitoring decisions.
10. To provide recommendations for improving fraud monitoring and future analytical capabilities.

1.5 Scope of the Project

The scope of this project includes the analysis of banking transaction records using data analytics techniques.

The project covers:

- Data inspection and validation.
- Data cleaning.
- Data quality checks.
- Feature engineering.
- Fraud distribution analysis.
- Transaction-level analysis.
- Customer-related analysis.
- Time-based analysis.
- Merchant and device analysis.
- Risk score development.
- Risk-level analysis.
- SQL-based analytical queries.
- Tableau dashboard development.
- Business insights and recommendations.

The project focuses primarily on historical transaction analytics and risk monitoring. It does not implement a machine learning fraud prediction model in the current version.

CHAPTER 2

DATASET DESCRIPTION

2.1 Dataset Overview

The dataset used for the project is a Bank Fraud Analytics Dataset containing banking transaction information.

The dataset was obtained from Dynamic Duniya and was provided in Excel format.

Dataset Characteristics

Dataset Name: Bank Fraud Analytics Dataset

Original Number of Records: 200,000

Original Number of Attributes: 24

File Format: Excel (.xlsx)

Target Variable: Is_Fraud

Fraud Label: 1 = Fraud, 0 = Genuine

Domain: Banking and Finance

Primary Task: Fraud Analytics and Risk Monitoring

2.2 Dataset Attributes

The original dataset contains the following 24 attributes:

Category	Attributes
Customer Information	Customer_ID, Customer_Name, Gender, Age, State, City
Account Information	Bank_Branch, Account_Type, Account_Balance
Transaction Information	Transaction_ID, Transaction_Date, Transaction_Time, Transaction_Amount, Transaction_Type
Merchant Information	Merchant_ID, Merchant_Category
Device Information	Transaction_Device, Device_Type, Transaction_Location
Fraud Information	Is_Fraud
Additional Information	Transaction_Currency, Customer_Contact, Transaction_Description, Customer_Email

2.3 Target Variable

The target variable used for fraud analysis is:

Is_Fraud

The variable contains two possible values:

- **0 – Genuine Transaction**
- **1 – Fraudulent Transaction**

The dataset contains:

- **189,912 Genuine Transactions**
- **10,088 Fraudulent Transactions**

Therefore, the overall fraud rate is approximately:

5.04%

This indicates that fraudulent transactions represent a relatively small proportion of the total transaction population.

2.4 Dataset Imbalance

The dataset contains considerably more genuine transactions than fraudulent transactions.

Approximately:

- **94.96%** of transactions are genuine.
- **5.04%** of transactions are fraudulent.

This imbalance is important because fraud analysis should not rely only on transaction counts. Fraud **rates and proportions** provide a more meaningful comparison between categories with different transaction volumes.

CHAPTER 3

TOOLS AND TECHNOLOGIES

USED

The project uses multiple data analytics and database technologies.

3.1 Python

Python was used for:

- Data loading.
- Data inspection.
- Data cleaning.
- Feature engineering.
- Statistical analysis.
- Exploratory data analysis.
- Visualization.

The major Python libraries used were:

- Pandas
- NumPy
- Matplotlib

3.2 Microsoft Excel

Excel was used as the primary format for storing and managing the dataset.

The cleaned and feature-engineered datasets were exported as Excel files for further analysis and documentation.

3.3 MySQL

MySQL was used for:

- Database creation.
- Data storage.

- Data validation.
- Analytical SQL queries.
- Aggregation.
- Fraud-rate calculations.
- Window functions.
- Common Table Expressions.
- SQL views.

A database named:

bank_fraud_analytics

was created.

3.4 Tableau

Tableau was used to develop an interactive dashboard for monitoring:

- Total transactions.
- Fraud transactions.
- Fraud rate.
- High-risk transactions.
- Fraud rate by transaction type.
- Fraud rate by device type.
- Fraud rate by merchant category.
- Fraud rate by state.
- Fraud rate by time of day.
- Fraud rate by risk level.

Dashboard filters were also incorporated to allow users to explore the data interactively.

CHAPTER 4

DATA CLEANING AND PREPROCESSING

4.1 Data Loading

The original dataset was loaded into Python using Pandas.

The dataset was read from:

Bank Fraud Analytics Dataset.xlsx

The structure, column names, dimensions, data types, first records, last records, and missing values were initially inspected.

4.2 Missing Value Analysis

A missing-value check was performed using Pandas.

The analysis checked every column for missing values.

Selected text-based fields with missing values were replaced with the value:

"Unknown"

The affected fields included:

- Customer_ID
- Customer_Name
- State
- Merchant_ID
- Transaction_Device
- Transaction_Location
- Customer_Contact
- Transaction_Description
- Customer_Email

This approach preserved the transaction records instead of unnecessarily removing rows.

4.3 Duplicate Record Check

Duplicate records were checked using:

```
banking.duplicated().sum()
```

The complete-row duplicate check resulted in:

0 duplicate rows

Therefore, no complete duplicate records were present after preprocessing.

4.4 Transaction ID Uniqueness Check

A separate validation was performed on Transaction_ID.

The analysis identified:

- Total Transaction IDs: **200,000**
- Unique Transaction IDs: **199,999**
- Duplicate Transaction IDs: **1**

The duplicated Transaction ID was investigated individually rather than automatically deleting one of the records.

The duplicate Transaction ID was:

b16e585d-31a8-4324-96e5-32f4c3d22cde

The comparison showed that the records were not identical and differed in fields including:

- Transaction_Type
- Customer_Contact

Therefore, the duplicate Transaction ID was treated as a data-quality issue requiring investigation, rather than being treated as a duplicate complete transaction.

4.5 Invalid Value Validation

Several validation checks were performed.

The following conditions were checked:

Age

The valid age range was defined as 18–100.

Invalid Age: 0

Transaction Amount

Negative transaction amounts were checked.

Invalid Transaction Amount: 0

Account Balance

Negative account balances were checked.

Invalid Account Balance: 0

Fraud Label

The Is_Fraud column was checked to ensure that values were restricted to 0 and 1.

Invalid Is_Fraud Values: 0

Transaction Currency

The transaction currency field was checked for missing values.

4.6 Data Type Validation

The data types of the dataset columns were examined using Pandas.

The transaction date was handled as a date/time variable, while transaction time was converted into a proper datetime representation for extracting the transaction hour.

This enabled additional time-based features to be created.

CHAPTER 5

FEATURE ENGINEERING

Feature engineering was performed to create additional analytical variables from the existing transaction data.

The following features were created:

1. Hour
2. Age_Group
3. Day_Name
4. month_name
5. Transaction_Size
6. Balance
7. Time_of_Day
8. Balance_Utilization
9. Risk_Score
10. Risk_Level

5.1 Hour

The transaction time was converted into a datetime format.

The hour component was extracted to create the:

Hour

feature.

This feature allows fraud patterns to be examined at different hours of the day.

5.2 Age Group

Customer ages were divided into four analytical groups:

Age Range	Age Group
18–25	Young
26–40	Adult
41–55	Middle Age
56–70	Senior

This categorization simplifies age-based fraud analysis.

5.3 Day Name

The transaction date was used to derive:

Day_Name

The feature identifies the day of the week on which each transaction occurred.

The categories include:

- Monday
- Tuesday
- Wednesday
- Thursday
- Friday
- Saturday
- Sunday

5.4 Month Name

The month was extracted from the transaction date to create:

month_name

This feature supports monthly fraud analysis.

The available transaction data belongs to January, so monthly trend interpretation should be considered within the available dataset period.

5.5 Transaction Size

Transaction amounts were classified into four categories:

Transaction Amount	Transaction Size
Less than 25,000	Low
25,000–49,999	Medium
50,000–74,999	High
75,000 and above	Very High

This feature allows fraud behaviour to be compared across transaction-value categories.

5.6 Balance Category

Account balances were divided into four categories:

Account Balance	Balance Category
Less than 25,000	Low
25,000–49,999	Medium
50,000–74,999	High
75,000 and above	Very High

The resulting feature was named:

Balance

5.7 Time of Day

The transaction hour was grouped into four periods:

Hour	Time of Day
00:00–05:59	Late Night
06:00–11:59	Morning
12:00–17:59	Afternoon
18:00–23:59	Evening

This feature is particularly useful for identifying unusual transaction timing.

5.8 Balance Utilization

A new feature called:

Balance_Utilization

was calculated using:

$$\text{Balance Utilization} = \text{Transaction Amount} / \text{Account Balance}$$

This metric represents the transaction amount relative to the available account balance.

Higher values may indicate transactions that are large relative to the account balance.

Validation was also performed to check:

- Minimum value.
- Maximum value.
- Missing values.
- Infinite values.
- Negative values.

5.9 Rule-Based Risk Score

A rule-based heuristic risk score was developed to support transaction risk monitoring.

The score is based on predefined analytical rules.

The following weights were assigned:

Risk Indicator	Score
Very High Transaction Size	+30
High Transaction Size	+20
Late Night Transaction	+20
Very High Balance	+15
Balance Utilization > 2	+35

The risk score is cumulative. Therefore, a transaction can receive multiple risk points if it satisfies multiple conditions.

Important Note

The Risk Score is an **analyst-defined heuristic scoring system**.

It is:

- Not generated by machine learning.
- Not trained using historical data.
- Not a predictive ML probability.
- Based on predefined business rules.

5.10 Risk Level

Based on the Risk Score, transactions were classified into three risk categories:

Risk Score	Risk Level
0–20	Low Risk
21–50	Medium Risk
Above 50	High Risk

The resulting distribution was:

- **Low Risk:** 101,589 transactions
- **Medium Risk:** 54,676 transactions
- **High Risk:** 43,735 transactions

5.11 Risk Score Validation

The relationship between the rule-based risk level and actual fraud was evaluated using fraud rates.

The results were:

Risk Level	Total Transactions	Fraud Transactions	Fraud Rate
Low Risk	101,589	3,435	3.38%
Medium Risk	54,676	3,147	5.76%
High Risk	43,735	3,506	8.02%

The results show that the fraud rate increases from Low Risk to Medium Risk and then to High Risk.

This provides evidence that the manually defined risk scoring framework has useful analytical value for risk monitoring.

5.12 Removal of Personal Information

For the analytical layer, the following personally identifiable information fields were removed:

- Customer_Name
- Customer_Contact
- Customer_Email

These fields were not required for aggregate fraud analysis or dashboard reporting.

The removal reduces unnecessary exposure of personal information within the analytical dataset.

CHAPTER 6

EXPLORATORY DATA

ANALYSIS

6.1 Fraud Distribution

The dataset contains:

- **189,912 Genuine Transactions**
- **10,088 Fraudulent Transactions**

The overall fraud rate is:

5.04%

This indicates that fraudulent transactions form a minority of the total transactions.

However, even a relatively small fraud percentage can represent significant risk when transaction volumes are large.

6.2 Gender Distribution

The dataset contains:

- Male: **100,452**
- Female: **99,548**

The distribution is almost balanced.

Therefore, the dataset does not show a major imbalance between male and female records in terms of overall transaction representation.

6.3 Account Type Distribution

The dataset contains three account types:

- Checking
- Savings

- Business

The transaction volumes across the three account types are approximately balanced.

Further fraud-rate analysis was performed to determine whether account type is associated with different levels of fraud.

6.4 Transaction Type Analysis

The transaction types included:

- Credit
- Debit
- Transfer
- Withdrawal
- Bill Payment

Fraud rates were calculated instead of relying only on fraud counts.

The Tableau dashboard showed the following fraud rates:

Transaction Type	Fraud Rate
Transfer	5.90%
Withdrawal	5.25%
Debit	4.86%
Credit	4.69%
Bill Payment	4.51%

Observation

Transfer transactions recorded the highest fraud rate at approximately 5.90%.

Bill Payment transactions recorded the lowest fraud rate among the five transaction types at approximately 4.51%.

This indicates that transfer transactions deserve comparatively greater attention during fraud monitoring.

6.5 Merchant Category Analysis

The merchant categories analyzed were:

- Electronics
- Entertainment
- Health
- Clothing
- Groceries
- Restaurant

The Tableau dashboard showed:

Merchant Category	Fraud Rate
Electronics	6.05%
Entertainment	5.80%
Health	4.78%
Clothing	4.68%
Groceries	4.66%
Restaurant	4.26%

Observation

Electronics had the highest fraud rate at approximately 6.05%.

Restaurant transactions had the lowest fraud rate at approximately 4.26%.

This suggests that merchant category can be useful as one of the dimensions for fraud monitoring.

6.6 Device Type Analysis

The dataset contains four device types:

- Mobile
- POS
- Desktop
- ATM

The dashboard showed the following fraud rates:

Device Type	Fraud Rate
Mobile	7.33%
POS	5.05%
Desktop	4.47%
ATM	3.23%

Observation

Mobile transactions had the highest fraud rate at approximately 7.33%.

ATM transactions had the lowest fraud rate at approximately 3.23%.

This indicates that mobile transactions represent an important area for enhanced fraud monitoring.

6.7 Time of Day Analysis

Fraud rates were calculated across:

- Late Night
- Morning
- Afternoon
- Evening

The dashboard showed:

Time of Day	Fraud Rate
Late Night	13.01%
Afternoon	2.08%
Morning	2.07%
Evening	2.02%

Observation

Late Night transactions showed a substantially higher fraud rate of approximately 13.01%.

The fraud rates during Morning, Afternoon, and Evening were approximately 2%.

This is one of the strongest patterns identified in the analysis.

Late-night transactions may therefore be considered an important monitoring condition, particularly when combined with other risk indicators.

6.8 Risk Level Analysis

The risk-level analysis showed:

Risk Level	Fraud Rate
High Risk	8.02%
Medium Risk	5.76%
Low Risk	3.38%

Observation

High-risk transactions recorded the highest fraud rate.

The fraud rate increased from:

3.38% → 5.76% → 8.02%

as the risk category increased from Low Risk to High Risk.

This supports the usefulness of the rule-based risk scoring framework as a monitoring mechanism.

6.9 State-wise Fraud Analysis

Fraud rates were calculated for individual states.

The analysis identified differences in fraud rates between states.

The Tableau dashboard displayed the highest-performing states based on fraud rate, with **Tamil Nadu** and **Lakshadweep** appearing among the highest-rate states in the displayed dashboard.

The analysis demonstrates that geographic location can also be considered as a dimension for fraud monitoring.

However, state-level differences should be interpreted together with transaction volume because a high fraud rate from a relatively small number of transactions may not represent the same operational risk as a high fraud rate from a large transaction population.

6.10 Age Group Analysis

Customer age was categorized into:

- Young
- Adult
- Middle Age
- Senior

Fraud rates were calculated for each age group.

This analysis helps determine whether fraudulent transactions are distributed differently across customer age segments.

Age should be treated as a supporting analytical factor rather than a standalone fraud indicator.

6.11 Transaction Amount Analysis

Transaction amounts were examined using:

- Histogram.
- Fraud vs non-fraud box plot.
- Transaction size categories.

The transaction-size feature divided transactions into Low, Medium, High, and Very High categories.

The analysis helps determine whether fraudulent transactions differ in monetary value from genuine transactions.

6.12 Account Balance Analysis

Account balance was analyzed using:

- Average balance.
- Minimum balance.
- Maximum balance.
- Fraud vs non-fraud comparisons.
- Balance categories.

This analysis was combined with transaction amount through the Balance Utilization feature.

6.13 Balance Utilization Analysis

Balance Utilization was calculated as:

Transaction Amount / Account Balance

The feature was compared between fraudulent and non-fraudulent transactions.

The analysis provides additional information beyond transaction amount alone because it considers the transaction relative to the customer's account balance.

6.14 Risk Score Analysis

The average Risk Score was compared between fraudulent and genuine transactions.

The project also examined fraud rate at different Risk Score values.

The Risk Score analysis provides a connection between the manually defined risk indicators and observed fraud behaviour.

6.15 Day-of-Week Analysis

Fraud rates were calculated for each day:

- Monday
- Tuesday
- Wednesday
- Thursday
- Friday
- Saturday
- Sunday

This analysis helps determine whether fraud behaviour changes according to the day of the week.

6.16 Fraud Trend Over Time

A daily fraud trend was generated using the transaction date.

The analysis plotted the number of fraudulent transactions across transaction dates.

Since the available dataset contains transactions from a limited period, the trend should be interpreted as a pattern within the available dataset rather than as a long-term banking fraud trend.

6.17 Correlation Analysis

A correlation matrix was created using numerical variables including:

- Age
- Transaction Amount
- Account Balance
- Balance Utilization
- Risk Score
- Is_Fraud

The correlation analysis provides an initial understanding of relationships between numerical variables and the fraud indicator.

Correlation does not establish causation. Therefore, correlated variables should be interpreted as associations rather than direct causes of fraudulent behaviour.

CHAPTER 7

SQL ANALYSIS

7.1 Database Creation

A MySQL database named:

bank_fraud_analytics

was created to store the feature-engineered transaction dataset.

A transactions table was created containing the original analytical fields and engineered features.

7.2 Data Import

The feature-engineered dataset was exported to CSV and imported into MySQL using Python and mysql.connector.

The import process used chunk-based loading with a chunk size of 1,000 records.

This approach allowed the large dataset to be inserted into MySQL in manageable batches.

7.3 SQL Data Validation

SQL queries were used to validate:

- Total number of rows.
- Unique Transaction IDs.
- Duplicate Transaction IDs.
- Missing values.
- Fraud distribution.
- Fraud percentages.

This provided a second layer of validation after Python-based preprocessing.

7.4 Fraud Analysis Using SQL

SQL aggregation functions such as:

- COUNT()
- SUM()
- AVG()
- MIN()
- MAX()
- ROUND()

were used to calculate fraud-related metrics.

For example, fraud rate was calculated as:

$$\text{Fraud Rate} = \text{Fraudulent Transactions} / \text{Total Transactions} \times 100$$

This calculation was used across transaction types, merchant categories, devices, states, account types, time periods, and risk levels.

7.5 Common Table Expression

A Common Table Expression (CTE) was used for monthly fraud trend analysis.

The CTE grouped transactions by year and month and calculated:

- Total transactions.
- Fraud transactions.
- Fraud rate.

This demonstrates the use of structured SQL querying for time-based analysis.

7.6 CASE WHEN

SQL CASE WHEN was used to recreate transaction size classification directly within SQL.

The categories were:

- Low
- Medium
- High
- Very High

This demonstrates how analytical features can also be generated through SQL logic.

7.7 RANK Function

The RANK() window function was used to rank states according to fraud rate.

This allows states to be ordered based on their relative fraud rates.

7.8 DENSE_RANK Function

The DENSE_RANK() window function was used to rank merchant categories according to fraud rate.

This demonstrates the application of window functions for analytical ranking.

7.9 Fraud Contribution Analysis

A window function was used to calculate the contribution of each transaction type to the total number of fraudulent transactions.

This provides a different perspective from fraud rate because it measures the share of all fraud transactions contributed by each transaction type.

7.10 High-Risk Segment Analysis

A combined analysis was performed using:

- Time of Day
- Device Type
- Transaction Size
- Risk Level

The SQL analysis identified combinations of these factors and calculated their:

- Total transactions.
- Fraud transactions.
- Fraud rate.

A minimum transaction count condition was applied to reduce the effect of extremely small groups.

CHAPTER 8

RISK MONITORING

FRAMEWORK

8.1 Overview

A rule-based risk monitoring framework was developed to identify transactions exhibiting predefined risk characteristics.

The framework does not predict fraud automatically. Instead, it assigns risk scores based on observable transaction characteristics.

8.2 Risk Indicators

The framework considers five main indicators:

1. Transaction Size
2. Time of Day
3. Account Balance Category
4. Balance Utilization
5. Combined risk conditions

Higher scores are assigned to conditions considered more significant within the analytical framework.

8.3 Risk Categories

The final categories are:

Low Risk

Risk Score ≤ 20

Medium Risk

Risk Score between 21 and 50

High Risk

Risk Score > 50

8.4 Risk Monitoring Result

The risk-level analysis produced:

Low Risk: 101,589 transactions

Medium Risk: 54,676 transactions

High Risk: 43,735 transactions

The corresponding fraud rates were:

Low Risk: 3.38%

Medium Risk: 5.76%

High Risk: 8.02%

The increasing fraud rate across the risk levels suggests that the rule-based framework can be useful for prioritizing transactions for monitoring.

CHAPTER 9

TABLEAU DASHBOARD

9.1 Dashboard Objective

An interactive Tableau dashboard was developed to provide a centralized view of fraud activity and risk patterns.

The dashboard is designed to support quick monitoring and comparison of fraud-related metrics.

9.2 Key Performance Indicators

The dashboard contains the following KPIs:

Total Transactions

200,000

Fraud Transactions

10,088

Fraud Rate

5.04%

High Risk Transactions

43,735

These KPIs provide an immediate overview of the transaction and fraud situation.

9.3 Dashboard Visualizations

The dashboard contains:

1. Fraud Rate by Transaction Type

2. Fraud Rate by Device Type
3. Fraud Rate by Risk Level
4. Fraud Rate by Merchant Category
5. Fraud Rate by State
6. Fraud by Time of Day

9.4 Dashboard Filters

Interactive filters were incorporated to allow users to analyze specific segments.

The dashboard includes filters such as:

- Transaction Date
- Transaction Type
- Device Type
- Account Type
- Risk Level

These filters allow users to explore fraud rates under different conditions.

9.5 Dashboard Findings

The dashboard provides several important findings:

- Overall fraud rate is approximately **5.04%**.
- Transfer transactions have the highest fraud rate among transaction types at approximately **5.90%**.
- Mobile transactions have the highest device-level fraud rate at approximately **7.33%**.
- Electronics has the highest merchant-category fraud rate at approximately **6.05%**.

- Late Night transactions have a substantially higher fraud rate of approximately **13.01%**.
- High Risk transactions have a fraud rate of approximately **8.02%**.

These findings demonstrate how the dashboard can support fraud-monitoring activities.

CHAPTER 10

KEY INSIGHTS

Based on the complete analysis, the following are the major findings of the project:

Insight 1: Fraud represents approximately 5.04% of transactions

Out of 200,000 transactions, 10,088 were identified as fraudulent.

Although fraud represents a minority of transactions, the volume is sufficiently large to require systematic monitoring.

Insight 2: Late Night transactions show a significantly higher fraud rate

The Late Night category recorded a fraud rate of approximately **13.01%**, substantially higher than Morning, Afternoon, and Evening.

This indicates that transaction timing can be an important fraud-monitoring factor.

Insight 3: Mobile transactions show the highest device-level fraud rate

Mobile transactions recorded a fraud rate of approximately **7.33%**, compared with 5.05% for POS, 4.47% for Desktop, and 3.23% for ATM.

This suggests that mobile transaction channels deserve additional monitoring attention.

Insight 4: Transfer transactions have the highest fraud rate among transaction types

Transfer transactions recorded a fraud rate of approximately **5.90%**, followed by Withdrawal at approximately **5.25%**.

This suggests that transaction type can be useful in identifying higher-risk transaction categories.

Insight 5: Electronics has the highest merchant-category fraud rate

Electronics transactions recorded a fraud rate of approximately **6.05%**, followed by Entertainment at approximately **5.80%**.

Merchant category can therefore be incorporated as an additional monitoring dimension.

Insight 6: Fraud rate increases with the rule-based risk level

Fraud rate increased from:

3.38% for Low Risk

to

5.76% for Medium Risk

and

8.02% for High Risk.

This indicates that the manually defined risk-scoring framework has a meaningful relationship with observed fraud rates.

Insight 7: Fraud monitoring should use rates rather than only fraud counts

Categories can contain different numbers of total transactions. Therefore, simply counting fraudulent transactions can produce misleading comparisons.

Fraud rate provides a more meaningful measure of the likelihood of fraud within each category.

CHAPTER 11

BUSINESS

RECOMMENDATIONS

Based on the findings, the following recommendations can be made for banking fraud-monitoring operations.

11.1 Strengthen Late-Night Transaction Monitoring

Since Late Night transactions showed a significantly higher fraud rate, banks could apply additional monitoring rules during late-night hours.

Transactions occurring during unusual hours could be subjected to additional verification depending on customer behaviour and transaction context.

11.2 Enhance Mobile Transaction Monitoring

Mobile transactions recorded the highest fraud rate among the analyzed device types.

Banks could therefore implement additional behavioural monitoring for mobile transactions, including device consistency, transaction frequency, transaction amount, and unusual transaction timing.

11.3 Monitor High-Risk Transaction Types

Transfer and Withdrawal transactions showed relatively higher fraud rates.

These transaction types could receive additional monitoring depending on transaction amount, customer history, location, and other available behavioural indicators.

11.4 Monitor High-Risk Merchant Categories

Electronics and Entertainment showed relatively higher fraud rates.

Banks and payment platforms could apply additional monitoring to transactions involving categories with consistently elevated fraud rates.

11.5 Use Risk Scores for Transaction Prioritization

The rule-based Risk Score can be used as a monitoring-support mechanism.

Instead of treating every transaction equally, transactions with higher risk scores could be prioritized for investigation.

11.6 Combine Multiple Risk Indicators

Fraud monitoring should not rely on a single variable.

For example, a transaction may deserve greater attention when multiple conditions occur together, such as:

Late Night + Mobile + Very High Transaction Size + High Risk

This combined approach can provide more meaningful risk segmentation.

11.7 Use Fraud Rate Alongside Fraud Count

Fraud counts should be considered together with total transaction volume.

Fraud rate helps identify categories where the proportion of fraudulent transactions is relatively high.

CHAPTER 12

LIMITATIONS OF THE STUDY

The project has several limitations.

12.1 Dataset Limitation

The analysis is based on a predefined banking transaction dataset. Therefore, the findings depend on the characteristics and quality of the available data.

12.2 Limited Customer History

The dataset primarily represents transaction records and does not provide a complete longitudinal customer history.

Therefore, long-term individual customer behaviour cannot be fully evaluated.

12.3 Rule-Based Risk Score

The Risk Score is manually defined using analyst-selected rules and weights.

The weights are not statistically learned from historical fraud data.

Therefore, the score should be considered a risk-monitoring heuristic rather than a predictive fraud probability.

12.4 No Machine Learning Model in the Current Version

The current implementation does not include a machine learning prediction model.

The project focuses on:

- Data analytics.
- Exploratory analysis.
- Feature engineering.

- SQL analysis.
- Rule-based risk scoring.
- Dashboard-based monitoring.

Machine learning can be considered as future work.

12.5 Correlation Does Not Establish Causation

The correlation analysis identifies relationships between variables but does not prove that a particular variable causes fraud.

12.6 Dataset Time Period

The available transaction data covers a limited time period, with the transaction month represented as January.

Therefore, the results should not be interpreted as a long-term seasonal fraud trend.

CHAPTER 13

FUTURE SCOPE

The project can be extended in several ways.

13.1 Machine Learning-Based Fraud Detection

A machine learning model can be developed using the engineered features.

Potential algorithms include:

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost
- Gradient Boosting

The model could generate fraud probabilities for new transactions.

13.2 Real-Time Fraud Detection

The project can be extended into a real-time fraud monitoring system where transactions are evaluated immediately after they occur.

13.3 Customer Behaviour Profiling

Future versions can analyze historical customer behaviour to identify unusual deviations from normal patterns.

For example:

- Unusual transaction amount.
- Unusual transaction location.
- Unusual transaction time.
- Unusual device.
- Sudden increase in transaction frequency.

13.4 Advanced Anomaly Detection

Unsupervised learning techniques can be introduced to identify unusual transactions even when fraud labels are unavailable.

Potential techniques include:

- Isolation Forest.
- Clustering.
- Autoencoders.

13.5 Automated Alerts

The risk-monitoring framework can be extended to generate alerts whenever transactions exceed predefined risk thresholds.

13.6 Advanced Dashboarding

The Tableau dashboard can be further enhanced with:

- Customer-level drill-down.
- Geographic maps.
- Fraud trend analysis.
- Risk-score distributions.
- Dynamic alerts.
- Advanced dashboard actions.
- Detailed transaction investigation views.

CHAPTER 14

CONCLUSION

The project “Data-Driven Bank Fraud Analytics and Risk Monitoring Using Transaction and Customer Behaviour Analysis” demonstrates how data analytics techniques can be applied to a large banking transaction dataset to identify fraud patterns and support risk monitoring.

The analysis was performed on 200,000 transactions, of which 10,088 transactions were fraudulent, resulting in an overall fraud rate of approximately 5.04%.

The project followed a structured analytical workflow involving data cleaning, validation, feature engineering, exploratory data analysis, SQL analysis, risk scoring, and dashboard development.

Several important fraud patterns were identified. Mobile transactions recorded a higher fraud rate than other device types, while Late Night transactions showed a substantially higher fraud rate. Transfer transactions recorded the highest fraud rate among transaction types, and Electronics recorded the highest fraud rate among the analyzed merchant categories.

A rule-based risk scoring framework was developed using transaction size, time of day, account balance, and balance utilization. The resulting risk levels showed increasing fraud rates from Low Risk (3.38%) to Medium Risk (5.76%) and High Risk (8.02%). This indicates that the heuristic risk framework can provide useful support for prioritizing transactions for monitoring.

SQL analysis further strengthened the project by providing database-level validation, aggregation, ranking, window-function analysis, and reusable analytical views.

Finally, the Tableau dashboard provides an interactive monitoring interface that enables users to examine fraud rates across transaction types, devices, merchant categories, states, time periods, and risk levels.

Overall, the project demonstrates that combining Python-based data analysis, feature engineering, SQL analytics, rule-based risk scoring, and Tableau visualization

can provide valuable insights into banking fraud behaviour and support data-driven fraud-monitoring decisions.

The current project provides an analytical foundation that can be further extended with machine learning, real-time transaction monitoring, anomaly detection, customer behavioural profiling, and automated fraud alerts.

APPENDIX A

SOFTWARE AND LIBRARIES

USED

Programming Language

Python

Python Libraries

- Pandas
- NumPy
- Matplotlib

Database

MySQL

Visualization Tool

Tableau Public

Data Storage Format

- Excel (.xlsx)
- CSV
- MySQL Database

APPENDIX B

ENGINEERED FEATURES

Feature	Description
Hour	Hour extracted from transaction time
Age_Group	Customer age classification
Day_Name	Day of the week
month_name	Month extracted from transaction date
Transaction_Size	Transaction amount classification
Balance	Account balance classification
Time_of_Day	Late Night, Morning, Afternoon, Evening
Balance_Utilization	Transaction Amount / Account Balance
Risk_Score	Analyst-defined heuristic risk score
Risk_Level	Low, Medium, High risk classification

APPENDIX C

RISK SCORING RULES

Condition	Weight
Very High Transaction Size	+30
High Transaction Size	+20
Late Night	+20
Very High Balance	+15
Balance Utilization > 2	+35

Risk Classification

Score	Risk Level
≤ 20	Low Risk
21–50	Medium Risk
> 50	High Risk

APPENDIX D

FINAL RISK SUMMARY

Risk Level	Transactions	Fraud Transactions	Fraud Rate
Low Risk	101,589	3,435	3.38%
Medium Risk	54,676	3,147	5.76%
High Risk	43,735	3,506	8.02%
Total	200,000	10,088	5.04%